Jay Doshi

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Education

B.Sc. in Computer Science & Engineering **GPA: 3.7/4.0**
University of Toledo, Toledo, OH Expected Graduation: December 2027

Professional Projects

MCP Chatbot-AI | Model Context Protocol, Ollama, FastAPI, Streamlit, GitHub OAuth **August 2025 – December 2025**

- Developed a local LLM chat app integrating Ollama models with Model Context Protocol for agentic, multi-step tool calling.
- Orchestrated a tool-calling loop of up to 10 iterations per query across 4 MCP tool categories, extensible via a GitHub MCP server.
- Implemented GitHub OAuth and 10 REST endpoints for chat, tool discovery, model switching, and server monitoring.

PolicyRAG Document Intelligence-AI | Python, RAG, GPT-4o, FAISS, FastAPI, Streamlit **December 2025 – January 2026**

- Built a RAG pipeline over 5 insurance policy documents (500+ pages), extracting text and tables from PDFs using pdfplumber.
- Indexed chunks in a FAISS vector store using OpenAI embeddings, reranking by cosine similarity to feed GPT-4o the top 5 clauses.
- Deployed a FastAPI endpoint with token auth answering batched queries in 40–50s; benchmarked retrieval latency and accuracy.

Web Memory Assistant-AI | FastAPI, Pinecone, Hugging Face, Groq, SQLite, Manifest V3 **August 2025 – December 2025**

- Architected a local-first browser memory system with a Manifest V3 extension and FastAPI backend for semantic page recall.
- Embedded 400-word chunks (50-word overlap) as 384-dimensional vectors in Pinecone, enabling semantic search over saved pages.
- Cut noisy captures with 100- and 50-word filter thresholds and automated page tagging through Groq-powered summarization.

DeepTruth-AI | Python, PyTorch, DistilBERT, Gemini API, AWS, GCP, MongoDB, React **January 2025 – March 2025**

- Fine-tuned DistilBERT on 9.5K+ labeled records, enabling dual-model scoring with 70% Gemini and 30% BERT weighting.
- Automated scheduled Transformer retraining and ML lifecycle management with AWS Lambda and CloudWatch cron workflows.
- Cleaned 9,500+ records with a Python ETL pipeline and automated validation checks, improving data ingestion accuracy.
- Won RocketHacks 2025 for building a scalable AI fact-checking product that processed data from 1,000+ users over 6 months.

Experience

Network Engineer Intern | Cenovus Energy, Oregon, OH **September 2026 – Present**

- Collaborated with infrastructure, OT, and process-control engineers to validate system integrations and support production cutovers.
- Monitored network and server health in SolarWinds NPM, flagging performance degradation before it escalated into outages.

Data Processing Engineer Intern | Cleveland-Cliffs, Coatesville, PA **May 2026 – August 2026**

- Engineered a classification-based anomaly detection tool analyzing live gauge data, reducing defective steel plate escapes by 30%.
- Engineered sensor-data features via ETL pipelines from legacy OpenVMS Alpha into SQL Server, cutting manual extraction 40%.
- Translated production data into Power BI dashboards on SQL views and stored procedures, accelerating decisions by 50%.

IT Support Technician Level 2 | University of Toledo, Toledo, OH **September 2025 – May 2026**

- Resolved 40+ monthly IT tickets by troubleshooting Windows, hardware, software, and network issues, minimizing downtime.
- Managed 200+ IT assets through inventory tracking, device deployment, secure decommissioning, and life-cycle documentation.

Undergraduate Research Assistant | University of Toledo, Toledo, OH **May 2025 – December 2025**

- Designed and evaluated a PPO-based reinforcement learning model for autonomous drone target navigation using ROS2-Gazebo.
- Increased control accuracy by 60% via statistical analysis of RMSE and overshoot in Pandas across 15+ hardware trials.
- Decreased target navigation error by 40% by tuning a custom PID-based flight model in a ROS2-Gazebo digital twin.

Technical Skills

AI & Machine Learning: Agentic-AI Workflow, LLM, RAG, MCP, LangChain, Transformer Models, Fine-Tuning, NLP, Deep Learning, Reinforcement Learning, MLOps

Programming Languages & Frameworks: Python, FastAPI, Streamlit, SQL, JavaScript, Java, C++, React, Django

ML Libraries & Data: PyTorch, Hugging Face, Pandas, NumPy, Matplotlib, Jupyter Notebook, Power BI Dashboards

Databases & Cloud: PostgreSQL, SQLite, Pinecone, FAISS, MongoDB, AWS (Lambda, S3, EC2), Google Cloud Platform

Developer Tools: Claude-Code (plugins, skills, hooks), Cursor, Git, GitHub Actions, Docker, Google Colab, VS Code